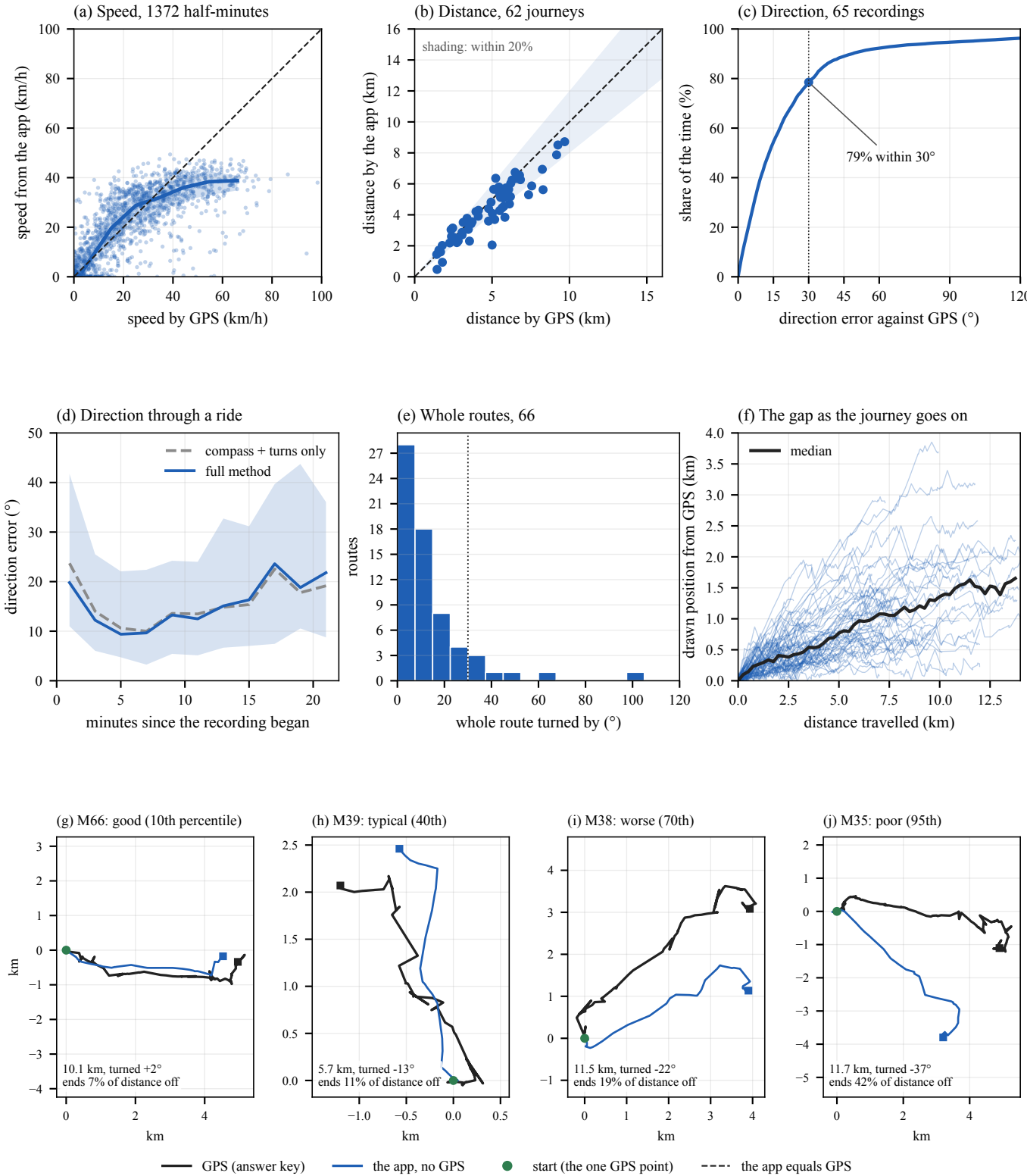

(a) Speed, 1372 half-minutes
speed from the app (km/h)
speed by GPS (km/h)
(b) Distance, 62 journeys
shading: within 20%
distance by the app (km)
distance by GPS (km)
(c) Direction, 65 recordings
share of the time (%)
79% within 30°
direction error against GPS (°)
(d) Direction through a ride
compass + turns only
full method
direction error (°)
minutes since the recording began
(e) Whole routes, 66
routes
whole route turned by (°)
(f) The gap as the journey goes on
median
drawn position from GPS (km)
distance travelled (km)
(g) M66: good (10th percentile)
10.1 km, turned +2°
ends 7% of distance off
(h) M39: typical (40th)
5.7 km, turned -13°
ends 11% of distance off
(i) M38: worse (70th)
11.5 km, turned -22°
ends 19% of distance off
(j) M35: poor (95th)
11.7 km, turned -37°
ends 42% of distance off
km
GPS (answer key)
the app, no GPS
start (the one GPS point)
the app equals GPS